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项目名称： “单原子催化剂”催化活性描述符的机器学习研究

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项目摘要

中文摘要:

从物理机理上深刻理解和精准构筑描述单原子催化剂 (SACs) 稳定性和催化效率的主导“描述符”，并以此为依据提升SACs的稳定性和优化其新奇物理化学特性是当今凝聚态物理、化学、材料等交叉学科中发展和制备经济、高效、功能型催化剂材料和器件的重点研究课题之一。本项目拟采用基于第一性原理为主的多尺度模拟计算与机器学习研究方法相结合，以CO的催化氧化为例，创新性地提出：通过厘清锚定单原子活性位点的载体 (Substrate) 的特征描述符S (比如功函数和还原性等) 与已知的金属单原子 (Atom) 的内秉属性描述符A (比如电负性等) 以及金属单原子与载体相互耦合 (Coupling) 的描述符C (比如配位数和结合能等) 之间的关联关系，从而构筑出描述SACs稳定性和催化效率的主导描述符D与S-A-C之间的明确对应关系，最终建立起SACs稳定性和催化性能的精准调控机理，为实验上制备新型高效的SACs提供重要的理论借鉴。

Abstract:

Constructing the dominant “descriptor” to describe the stability and catalytic efficiency of single-atom catalysts (SACs) from the physical mechanism point of view, enhancing their stabilities and optimizing their exotic physical and chemical properties are of the most fundamental and important research topics in the fields of condensed matter of physics, chemistry, and material science in establishing economic, highly efficient, and functional materials and devices. Here, taking CO oxidation as a prototypical example, we adopt multi-scale computational simulations combined with machine learning to systematically investigate and specifically establish the dominated descriptor (D) of the stability and reactivity of SACs. Specifically, we perform in-depth analysis on the relationship of the well-established descriptors of the supporting substrate (S), such as work function and reducibility, free-standing single atoms (A), such as electronegativity, and the descriptor that accounts for the coupling (C) of the deposited single atoms and the substrate, such as coordination number and binding energy, to unambiguously establish the new dominant descriptor D of the SACs systems in reflecting the stability and reactivity of SACs stabilized on the substrate, via figuring out the specific relationship of the D and S-A-C. Based on this, we establish effective modulation mechanism to achieve highly stable and excellently catalytic SACs. The present proposal is expected to provide valuable theoretical guidelines in developing new highly efficient SACs.

关键词 (用分号分开): 凝聚态化学; 第一性原理计算; 单原子催化剂; 掺杂; 机器学习

Keywords (separated by;): Condensed matter chemistry; First-principles calculations; Single-atom catalysts; Doping; Machine learning



结题摘要

中文摘要（对项目的背景、主要研究内容、重要结果、关键数据及其科学意义等做简单概述）：

本项目基于第一性原理密度泛函理论，系统研究还原性氧化物载体与过渡金属单原子的相互作用机制，构筑描述SACs稳定性和催化效率的主导“描述符”，建立SACs稳定性和催化性能的精准调控机理。

成果如下：1) 碱金属锚定无缺陷TiO₂(110) SACs：利用Li、Na、K原子在无缺陷表面实现Pd/Pt单原子的自组装形成AMA2 - O - Pd/Pt结构，CO氧化活性显著提升；发现Pt/Pd价态与催化活性高度相关，确立为催化活性描述符（JACS, submitted）。2) P掺杂PtCo纳米花海水电解制氢：在天然海水中仅需18.2 mV过电位即可达10 mA cm⁻²，稳定运行500小时；析氢电子结构调控提供新策略（ACS AMI, submitted）。3) 二维多铁性材料室温CO催化：Cr(h-fpyz)₂ SAC反应势垒低至0.325 eV，室温反应速率高达3.47×10⁶ s⁻¹，较单铁性衍生物提升6个数量级；揭示极化和自旋协同催化机理（Nat. Commun. 2025, 16, 1580）。4) 应变调控边缘氮掺杂石墨烯纳米带ORR：13 - 14%应变下过电位低至0.34 V，明确应变对活性位点电子结构与吸附行为的调控规律（Appl. Surf. Sci. 2025, 689, 162538）。5) TiO₂纳米管台阶位点负载FePc：台阶固有磁场诱发Fe 3d自旋翻转，增强Fe-O共价键，加速ORR质子化步骤，验证自旋翻转策略可降低O₂活化能垒（ACS Catal. 2024, 14, 7191）。6) p-block Sb/Bi自旋极化SACs：无缺陷TiO₂锚定Sb、Bi形成p-SAC，遵循Wigner自旋守恒，O₂活化与CO氧化决速步势垒0.6 eV，性能优于非磁性贵金属SACs（PCCP 2024, 26, 16459）。7) 动态受限单原子催化剂：多孔g-C₃N₄/MoS₂异质结限域单原子，反应时动态调控电荷与自旋，反应后返回稳定构型，实现高稳定性与高活性统一（Phys. Rev. B 2023, 107, 205421）。

本项目从电子结构、自旋态、应变、限域等多维度揭示了SACs稳定性与催化活性的微观调控机制，建立了描述符理论框架，为无缺陷基底高载量SACs、海水电解、室温高效催化、自旋催化及动态受限催化等前沿领域提供了系统的理论设计与调控策略。

Abstract (Brief description of research background, main methods, contributions, and research data):

This project based on first-principles density functional theory, and systematically investigates the interaction mechanisms between supports and transition atoms, constructs dominant “descriptors” for describing the stability and catalytic efficiency of SACs, and establishes precise regulation mechanisms for SAC catalytic performance.

The main findings are as follows: (a) Alkali-metal-anchored SACs on defect-free TiO₂(110): Using Li, Na, and K atoms, self-assembled stabilization of Pd/Pt single atoms is achieved on surfaces, forming an AMA2 - O - Pd/Pt structure with significantly enhanced CO oxidation activity. A strong correlation between the valence state of Pt/Pd and catalytic activity is revealed, establishing it as a descriptor for catalytic activity. (JACS, submitted). (b) P-doped PtCo nanoflowers for seawater electrolysis hydrogen production: P doping shifts the d-band center of Pt downward by 0.038 eV, optimizing hydrogen adsorption energy. In natural seawater, an overpotential of only 18.2 mV is required to achieve 10 mA cm⁻², with stable operation for 500 hours. This offers a new electronic structure regulation strategy for seawater electrolysis catalysts (ACS AMI, submitted). (c) 2D multiferroic materials for room-temperature CO catalysis: The Cr(h-fpyz)₂ single-atom catalyst exhibits a reaction barrier as low as 0.325 eV and a room-temperature reaction rate of 3.47×10⁶ s⁻¹, six orders of magnitude higher than its single-ferroic derivative Cr(pyz)₂. The synergistic catalysis mechanism of polarization and spin is revealed (Nat. Commun. 2025, 16, 1580). (d) Strain-engineered edge-nitrogen-doped graphene nanoribbons for ORR: Under 13 - 14% strain, the overpotential is as low as 0.34 V, clarifying the regulation of strain on the electronic structure and adsorption behavior of active sites (Appl. Surf. Sci. 2025, 689, 162538). (e) FePc anchored on TiO₂ nanotube step sites: The intrinsic magnetic field at TiO₂



atomic step sites induces spin flip of Fe 3d near the Fermi level, enhancing Fe-O covalent bonding and accelerating the ORR protonation step. This validates the spin-flip strategy for reducing the O₂ activation barrier (ACS Catal. 2024, 14, 7191). (f)p-block Sb/Bi spin-polarized SACs: Defect-free TiO₂ anchors Sb and Bi to form p-SACs, following Wigner spin conservation. The rate-determining step barrier for O₂ activation and CO oxidation is 0.6 eV, outperforming non-magnetic, low-spin noble metal SACs (PCCP 2024, 26, 16459). (f)Dynamically confined single-atom catalysts: Single atoms are confined in porous g-C₃N₄/MoS₂ heterojunctions, dynamically regulating charge and spin during the reaction and returning to a stable configuration post-reaction, achieving high activity (Phys. Rev. B 2023, 107, 205421).

关键词（用分号分开）： 单原子催化剂； 密度泛函理论； 描述符； 催化机理； 调控策略

Keywords (separated by;): Single-atom catalysts (SACs); Density functional theory (DFT); Descriptors; Catalytic mechanisms; Regulation strategies